

Anja Vogt

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EDUCATION

University of Washington

M.S. in Applied Mathematics

Seattle, WA

Sept 2024 – June 2025

- **GPA:** 3.9

- **Relevant Courses:** Scientific Computing, Computational Models in Biology, Applied Complex Analysis, Computational Methods for Data Analysis, Mathematical Analysis in Biology and Medicine, Dynamical Systems and Chaos, Analysis of Environmental and Ecological Data, Infectious Disease Dynamics: Models and Data, Methods for Partial Differential Equations

Case Western Reserve University

B.S. in Mathematics, Minor in History

Cleveland, OH

Aug 2020 – May 2024

- **GPA:** 4.0

- **Relevant Courses:** Applied Probability and Stochastic Processes for Biological Systems, Mathematical Analysis of Biological Models, Statistics for Science and Engineering, Differential Equations, Linear Algebra, Real Analysis I-II, Complex Analysis, Intro to Scientific Computing, Discrete Math, Number Theory, Abstract Algebra, Sets, Logic, and Categories, Intro to Advanced Math, Calculus I-III

DIS Abroad Copenhagen

Study Abroad

Copenhagen, Denmark

Summer 2022

- Climate Change and Glaciers, with a study tour to Iceland

RESEARCH AND PROJECTS

Atmospheric Chemistry and Mixing in the Convective Boundary Layer

Case Western Reserve University

Fall 2021 – Summer 2024

Cleveland, OH

- Advisor: Dr. David Gurarie, Department of Mathematics, Applied Mathematics, and Statistics
- Conducted a systematic analysis of various models of reactive mixing chemistry, including reaction kinetics (ODEs), reaction-diffusion systems (PDEs), and models accounting for turbulent mixing
- Focused on developing and improving the SOMCRUS (Second-Order Moment for Conserved and Reactive Unsteady Species) methodology to better understand and predict reactive chemical cycles through the atmosphere
- Summer 2023 research was supported by CWRU SOURCE (Undergraduate Research Office)

Machine Learning for Breast Cancer Image Classification

University of Washington

Jan – March 2025

Seattle, WA

- Built and evaluated machine learning classifiers using scikit-learn in Python to distinguish between benign and malignant tissue slices in histopathology images
- Trained models using a dataset of 7909 images, performed hyperparameter tuning, applied k-fold cross-validation to assess performance, compared testing accuracy across models

Modeling Measles Transmission from Historical Case Reports

University of Washington

March – June 2025

Seattle, WA

- Modeled measles outbreaks in Pennsylvania and West Virginia from 1930–1950 using historical weekly case reporting data
- Built and calibrated time-series SIR models in R; performed exploratory data analysis, fitted models to data, and analyzed transmission dynamics over time

Statistical Modeling of Lichen as Air Quality Bioindicators

University of Washington

March – June 2025

Seattle, WA

- Investigated the relationship between air pollutants (NO_x, SO₂) and lichen morphology through environmental data analysis using citizen science observation data
- Applied statistical inference techniques in R; fitted and interpreted generalized linear models (GLMs) to evaluate pollutant effects on lichen type distributions (fruticose, foliose, crustose)

Colorectal Cancer Modeling

University of Washington

Sept – Dec 2024

Seattle, WA

- Studied and extended a deterministic and stochastic model for cell population dynamics quantifying homeostasis in colonic crypts and its disruption in early tumorigenesis
- Evaluated model validity and biological implications, providing insights into cancer formation mechanisms
- Produced numerical simulations of models using MATLAB, implemented stochastic simulations using Gillespie's Exact Algorithm

Modeling the Spread of Schistosomiasis

Case Western Reserve University

Spring – Summer 2022

Cleveland, OH

- Advisor: Dr. David Gurarie, Department of Mathematics, Applied Mathematics, and Statistics
- Analyzed systems of differential equations to model the spread of schistosomiasis using the MacDonald system
- Focused on refining mathematical simulation models for disease transmission dynamics and control strategies

MENTORSHIP

Women in Applied Mathematics Mentorship (WAMM)

University of Washington

Spring 2025

Seattle, WA

- Mentorship program matching undergraduate STEM majors with female graduate students for quarter-long independent study and research project in applied mathematics
- Directed project development, introduced and taught relevant math topics, provided resources for undergraduates as they learn new material, facilitated weekly meetings, and prepared students for a final research presentation at the WAMM colloquium
- Designed and implemented a cancer modeling project using deterministic differential equations, stochastic simulations, and data-driven machine learning approaches to understand the dynamics of tumor formation under varying biological assumptions

TECHNICAL SKILLS

Programming Languages, Libraries, Tools, and Software

Experience with:

- Python, R, Wolfram Mathematica, MATLAB
- SciPy, Scikit-learn, PyTorch, Matplotlib libraries
- Visual Studio Code, RStudio, Jupyter Notebooks, LaTeX, Berkeley Madonna

Data Analysis

Experience with:

- Machine learning and neural networks for classification, regression, and dimensionality reduction
- Statistical methods: Principal Component Analysis, Linear Models, Linear Mixed Models, Generalized Linear Models, Generalized Linear Mixed Models, Hypothesis Testing, Bootstrapping and Resampling Techniques
- Mathematical Modeling: ODEs, PDEs, and stochastic models for mechanistic modeling; numerical methods and scientific computing
- Various forms of data including biomedical imaging data, citizen science observation data, air quality data, US natality data, infectious disease case reporting data

WORK EXPERIENCE

Teaching Assistant, Quantitative Science 292 (Calculus for Natural Systems II)

University of Washington, Center for Quantitative Science

Jan 2025 – June 2025

Seattle, WA

- Facilitated help sessions during two weekly class sections and held office hours to support student learning and problem-solving
- Graded assignments and exams, provided feedback on calculus techniques in ecological systems modeling
- Collaborated with the instructor on course materials, exam preparation, and in-class activities

Teaching Assistant, MATH 124 (Calculus I)

University of Washington, Department of Mathematics

Sept 2024 – Dec 2024

Seattle, WA

- Taught two sections of 30 students through weekly 80-minute worksheet sessions and 50-minute quiz/review sessions
- Held office hours, graded exams, and supported students with conceptual understanding and problem-solving techniques
- Acted as primary point of contact for students and liaison between students and the course instructor

Spoken English Language Partners (SELP) Tutor

Aug 2021 – May 2024

*Case Western Reserve University**Cleveland, OH*

- Worked one-on-one with international graduate students to build conversational fluency, pronunciation, and vocabulary
- Helped students improve classroom participation, presentation delivery, and confidence in academic settings
- Supported by CWRU Graduate Studies to promote cross-cultural communication and academic success

Math Tutor

2018 – 2024

*Private Tutoring**Various Locations/Online*

- Provided individualized tutoring in math topics from basic arithmetic to calculus for K–12 students with diverse learning styles and educational needs
- Helped students build foundational skills, improve grades, and increase confidence in problem-solving

Alumnae Fundraising Intern

June – August 2021

*Women and Girls Foundation of Southwest PA/GirlGov**Pittsburgh, PA*

- Conducted alumni research and networking, managed fundraising database
- Collaborated on communications and marketing, developed newsletters and social media posts
- Planned alumni summit event with team of organizers and program directors from WGF
- As program participant: Co-leader of Health and Environment Committee and Member of Leadership Council

CERTIFICATIONS

The Erdős Institute

May – August 2025

Data Science Boot Camp

- Completed an end-to-end data science project by defining a problem in quantitative terms, identifying KPIs, preparing and transforming data, and building supervised learning models for regression and classification.
- Built machine learning models to predict low birth weight and eclampsia from U.S. natality data, examining air quality as a predictive factor and prioritizing low false negative rates.

Wolfram Mathematica Certification

July 2023

Daily Study Group: Solving ODEs and PDEs with Wolfram Language

- Course on solving ODEs and PDEs using Wolfram Language, focusing on symbolic (DSolve) and numerical (NDSolve, FEM) methods for boundary value and eigenvalue problems.

HONORS AND AWARDS

- Tung Family Award (UW AMATH, Awarded Spring 2025)
- The Max Morris Prize (CWRU, Awarded Spring 2024)
- The John Schoff Millis Award (CWRU, Awarded Spring 2024)
- The Webster Godman Simon Mathematics Award (CWRU, Awarded Spring 2023)
- CWRU Dean's High Honor List (2020 - 2024)

MEMBERSHIPS

- Association for Women in Mathematics (AWM) – Member, CWRU and UW chapters (since 2022)
- Society for Industrial and Applied Mathematics (SIAM) – Member, UW chapter (since 2024)